

Advanced Visuals in Tableau

Assignment — Completed Answers

Dataset: Healthcare Dataset — a simulated dataset of 1,200 patient records built to match the assignment's fields: Patient_ID, Department, Region, Diagnosis, Month, Gender, Age, Treatment_Cost, and Satisfaction_Score, spanning Jan–Dec 2024 across 6 departments and 4 regions.

Note: Since Tableau Desktop isn't available in this environment, each advanced visual below was built programmatically (pandas + seaborn/matplotlib) using the same fields, aggregations, and chart logic a Tableau worksheet would use, and the Tableau shelf/build steps are described for each one.

Question 1: Heatmap — Treatment Cost by Department and Region

Tableau build steps: Place **Department** on Rows and **Region** on Columns. Drag **Treatment_Cost** onto Color (aggregation = Average). Set Marks type to 'Square'. Optionally drag Treatment_Cost onto Label as well to show the value inside each cell.

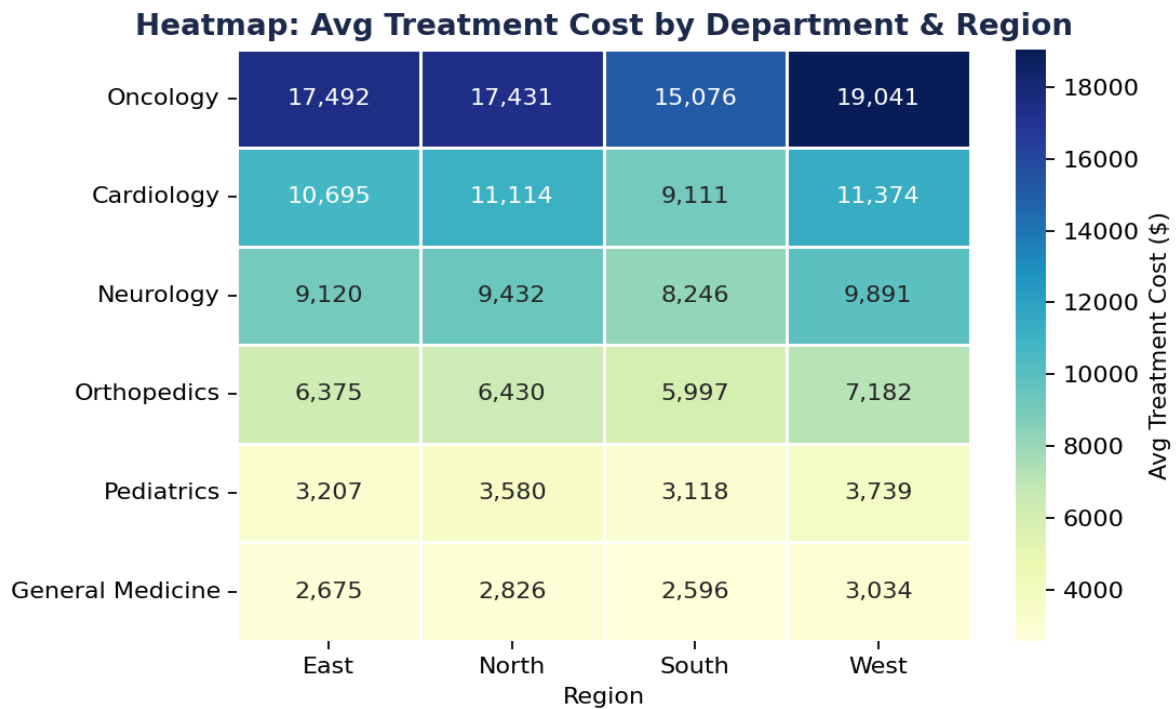


Figure 1: Average Treatment Cost (\$) by Department and Region

Insight: **Oncology** has by far the highest average treatment cost in every region (~\$15,000–\$19,000), with the **West** region consistently the most expensive across nearly all departments. **General Medicine** is the least costly department overall, and **South** is generally the least expensive region — useful for regional budgeting and cost-control initiatives.

Question 2: Highlight Table — Satisfaction Score by Department

Tableau build steps: Place **Department** on Rows, drag **Satisfaction_Score** (Average) onto the Text/Label shelf, then also drag it onto Color. Change the Marks type to 'Square' with a diverging color palette (e.g., Red-Yellow-Green) — this produces a single-column table where cell color and the numeric label both highlight performance.

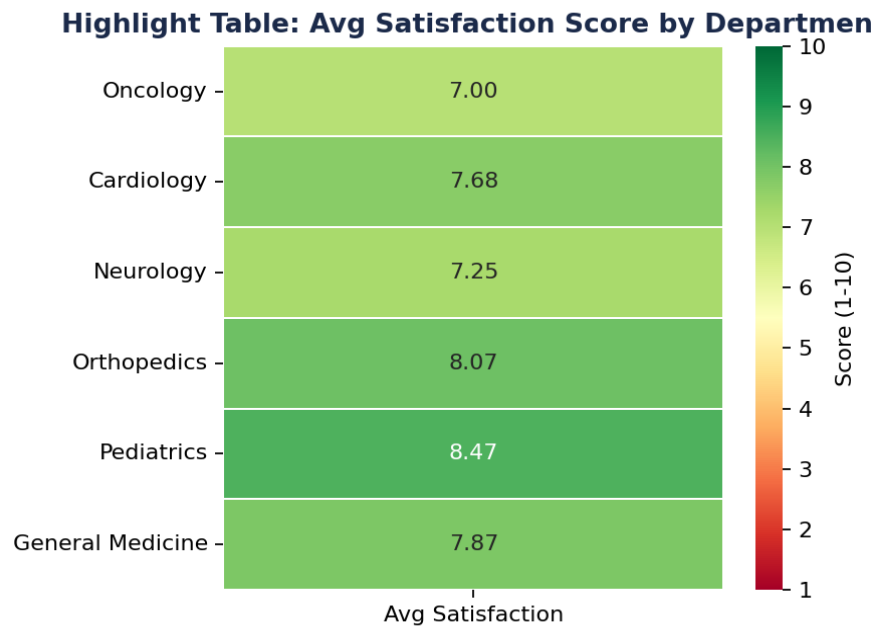


Figure 2: Average Satisfaction Score by Department (1–10 scale)

Insight: **Pediatrics** has the highest average satisfaction (8.47), followed by Orthopedics (8.07) and General Medicine (7.87). **Oncology** has the lowest satisfaction score (7.00) despite being the highest-cost department — a strong signal that patient experience in Oncology needs targeted improvement even though (or perhaps because) treatment there is the most expensive and complex.

Question 3: Bump Chart — Department Ranking by Treatment Cost Over Months

Tableau build steps: Create a calculated field '**Cost Rank**' using `RANK(SUM([Treatment_Cost]))` computed using Department, addressed/partitioned by Month. Place **Month** on Columns, the Rank calculation on Rows (reversed axis), and **Department** on Color/Detail. Use a Line mark with circles at each point — this is the standard Tableau bump-chart recipe.

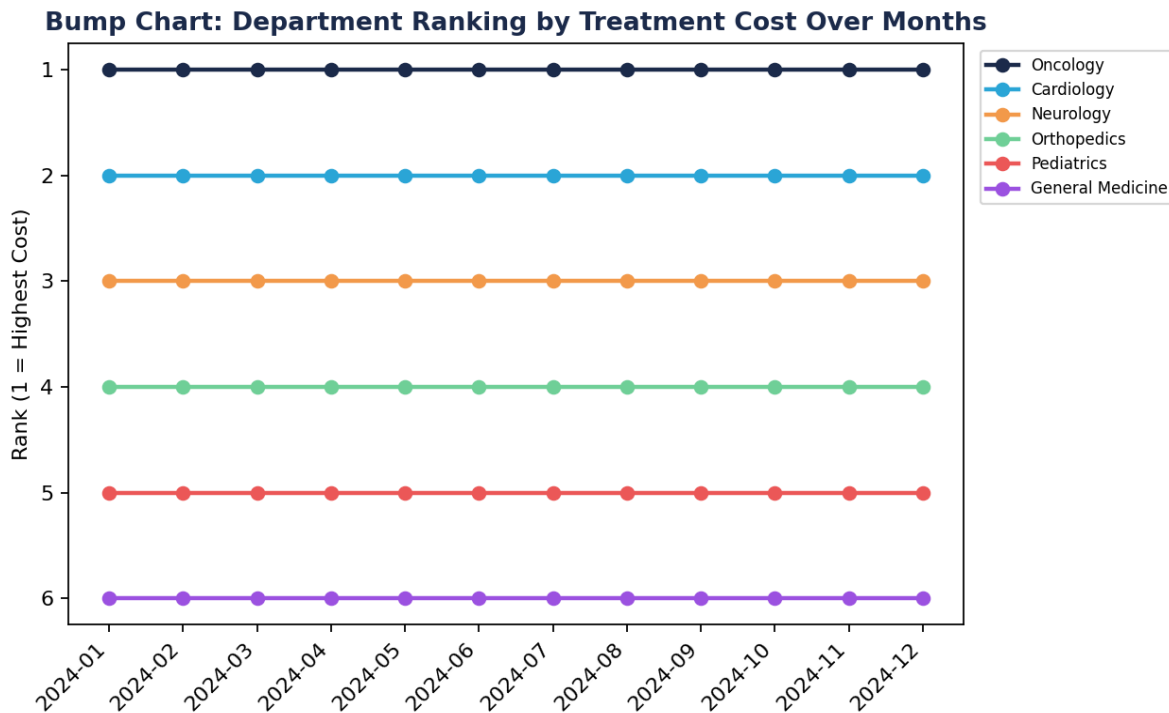


Figure 3: Department Ranking by Monthly Treatment Cost (Rank 1 = highest cost)

Insight: **Oncology** holds the #1 (highest-cost) rank in essentially every month, with **Cardiology** stable at #2 and **General Medicine** consistently ranked last (#6). The ranking order is quite stable month-to-month, showing cost hierarchy across departments is structural rather than seasonal — useful for setting department-specific budget tiers.

Question 4: Butterfly Chart — Male vs Female Patient Count by Department

Tableau build steps: Place **Department** on Rows. Create a calculated field that returns - [Patient Count] for Male and the plain count for Female, or place COUNT of patients on Columns and drag **Gender** onto Color, then edit the axis for the Male pane to be reversed. The classic approach uses two mirrored bar charts (Male negative, Female positive) combined into a single 'butterfly' view with a shared central axis.

Butterfly Chart: Male vs Female Patient Count by Department

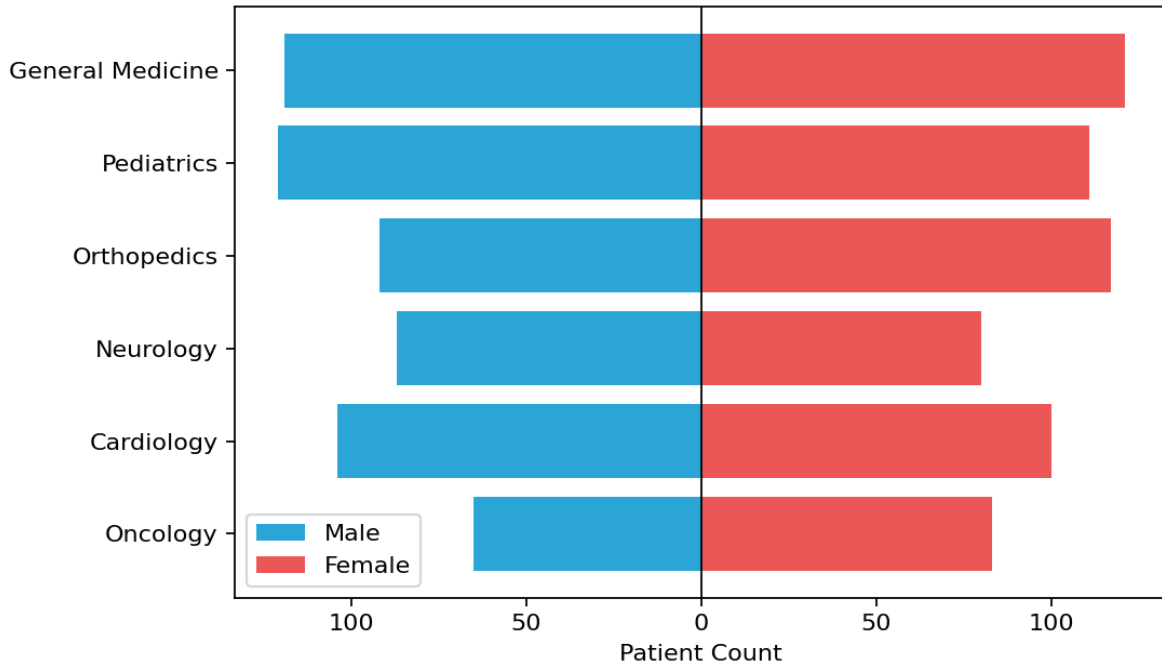


Figure 4: Male (left) vs Female (right) Patient Count by Department

Insight: Most departments are fairly balanced by gender. **Orthopedics** and **Pediatrics** skew slightly female (117 vs 92, and 111 vs 121 respectively), while **Neurology** and **Cardiology** have a marginally higher male patient count. **Oncology** shows the largest gap (83 female vs 65 male), which could inform targeted screening or outreach programs by gender within each department.

Question 5: Dual Axis Chart — Average Age vs Treatment Cost by Department

Tableau build steps: Place **Department** on Columns. Drag **AVG(Age)** onto Rows (Bar mark) and **AVG(Treatment_Cost)** onto Rows again (Line mark) to create a second axis. Right-click the second axis pill → 'Dual Axis', then 'Synchronize Axis' if a shared scale is desired (here the scales differ, so axes are kept independent — one for age in years, one for cost in dollars).

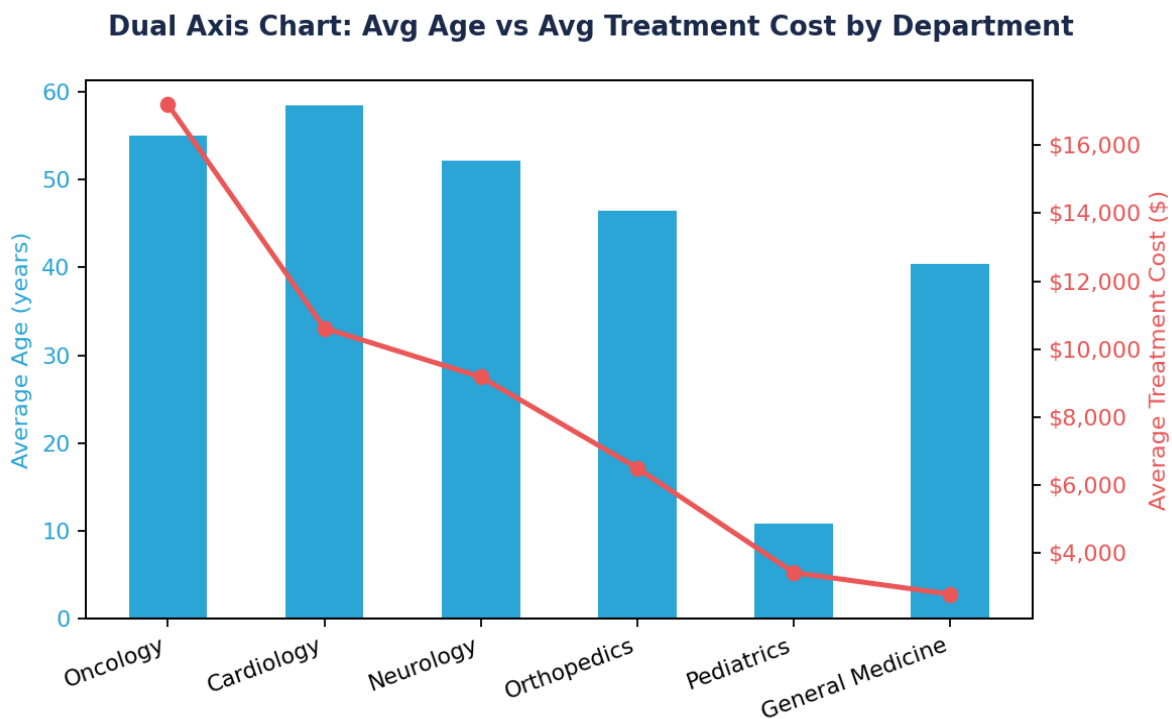


Figure 5: Average Age (bars) vs Average Treatment Cost (line) by Department

Insight: Treatment cost tracks patient age reasonably closely for adult-dominant departments — Cardiology and Oncology both skew toward older patients (avg. age 55–58) and carry the highest costs. **Pediatrics** is the clear exception: despite the youngest patients (avg. age ~11), its costs (\$3,428 avg) are still higher than General Medicine, showing age alone doesn't fully explain cost — diagnosis complexity matters too.

Question 6: Sunburst Chart — Department → Diagnosis Hierarchy by Treatment Cost

Tableau build steps: Tableau doesn't have a native sunburst mark, so this is built by nesting two Pie charts of different radii on the same worksheet (a common Tableau 'sunburst' workaround): an inner pie sized by SUM(Treatment_Cost) at the **Department** level, and an outer ring sized by SUM(Treatment_Cost) at the **Department** → **Diagnosis** level, both using the same color per department (with the outer ring using a lighter shade), aligned and overlaid using a dual-axis 'donut' technique.

Sunburst Chart: Department → Diagnosis by Treatment Cost

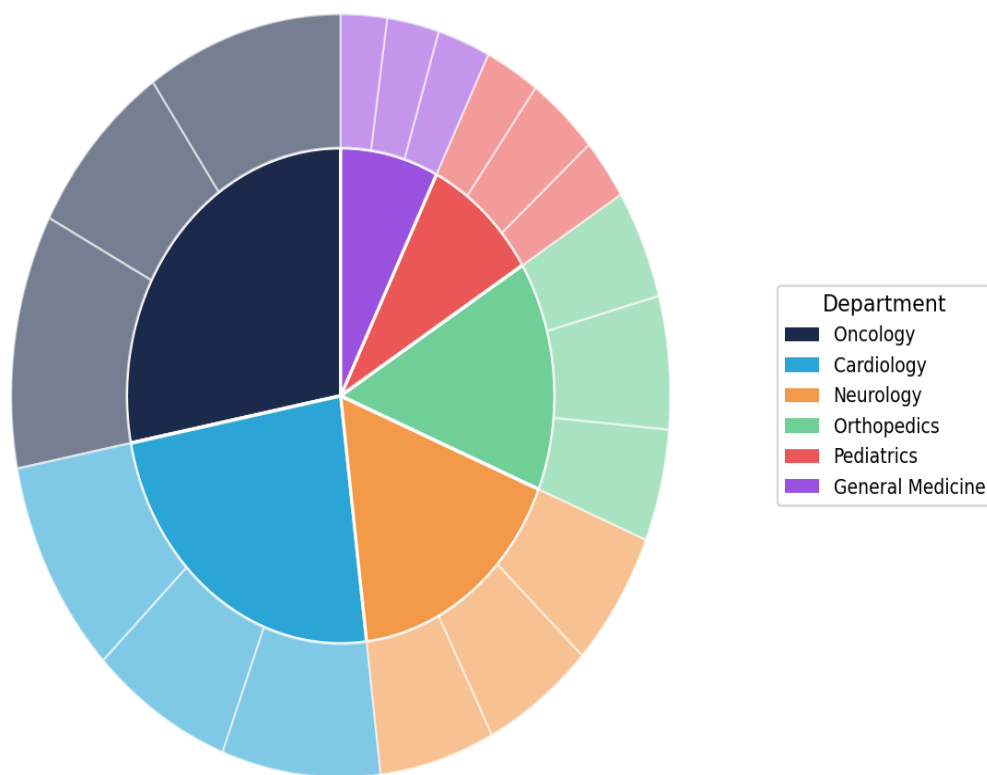


Figure 6: Department → Diagnosis Hierarchy Sized by Total Treatment Cost

Insight: **Oncology** occupies the largest share of the inner ring, and within it, **Lung Cancer** (~\$976K total cost) is the single largest wedge in the entire chart — larger than the total cost of any diagnosis in any other department. This immediately tells management that Lung Cancer treatment is the single biggest cost driver in the hospital system and warrants focused cost-management and resource planning.

Question 7: Dual Axis Chart — Treatment Cost vs Satisfaction Score by Department

Tableau build steps: Same technique as Question 5 — Department on Columns, AVG(Treatment_Cost) as a Bar on Rows, AVG(Satisfaction_Score) as a Line on Rows, combined via 'Dual Axis'. The satisfaction axis is fixed to a 0–10 range to keep the scale meaningful and comparable across departments.

Dual Axis Chart: Treatment Cost vs Satisfaction Score by Department

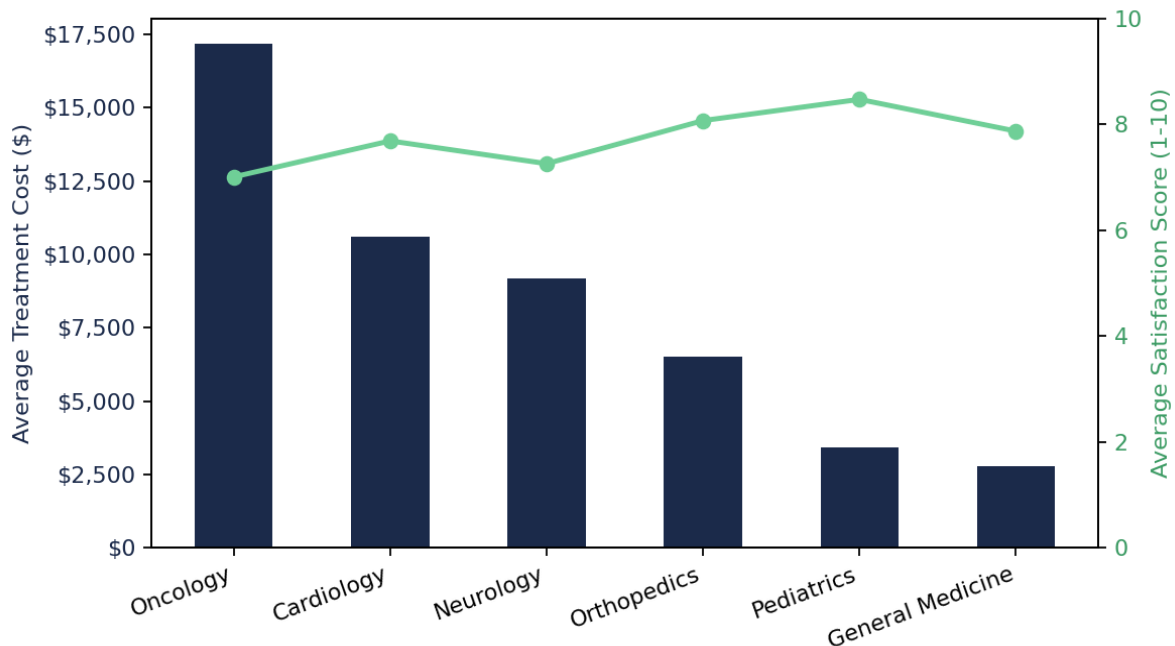


Figure 7: Average Treatment Cost (bars) vs Average Satisfaction Score (line) by Department

Insight: There's a visible inverse-like pattern at the extremes: **Oncology** has the highest cost but the lowest satisfaction (7.00), while **Pediatrics** has one of the lowest costs but the highest satisfaction (8.47). This suggests that higher spending does not automatically translate into a better patient experience — Oncology may need process/communication improvements independent of its clinical cost structure.

Question 8: Executive Dashboard — Combined View

Tableau build steps: Create a new Dashboard and drag the four worksheets (Heatmap, Dual Axis Age/Cost, Sunburst, Butterfly) onto a 2×2 tile/floating layout. Add **Department** and **Region** as dashboard filters set to 'Apply to Worksheets → All Using This Data Source', so choosing a department or region on any one chart cross-filters all four simultaneously. A dashboard title and a couple of KPI 'Big Number' tiles (e.g., Total Patients, Total Treatment Cost) would typically sit above the grid.

Healthcare Executive Dashboard (Filters: Department | Region | Month)

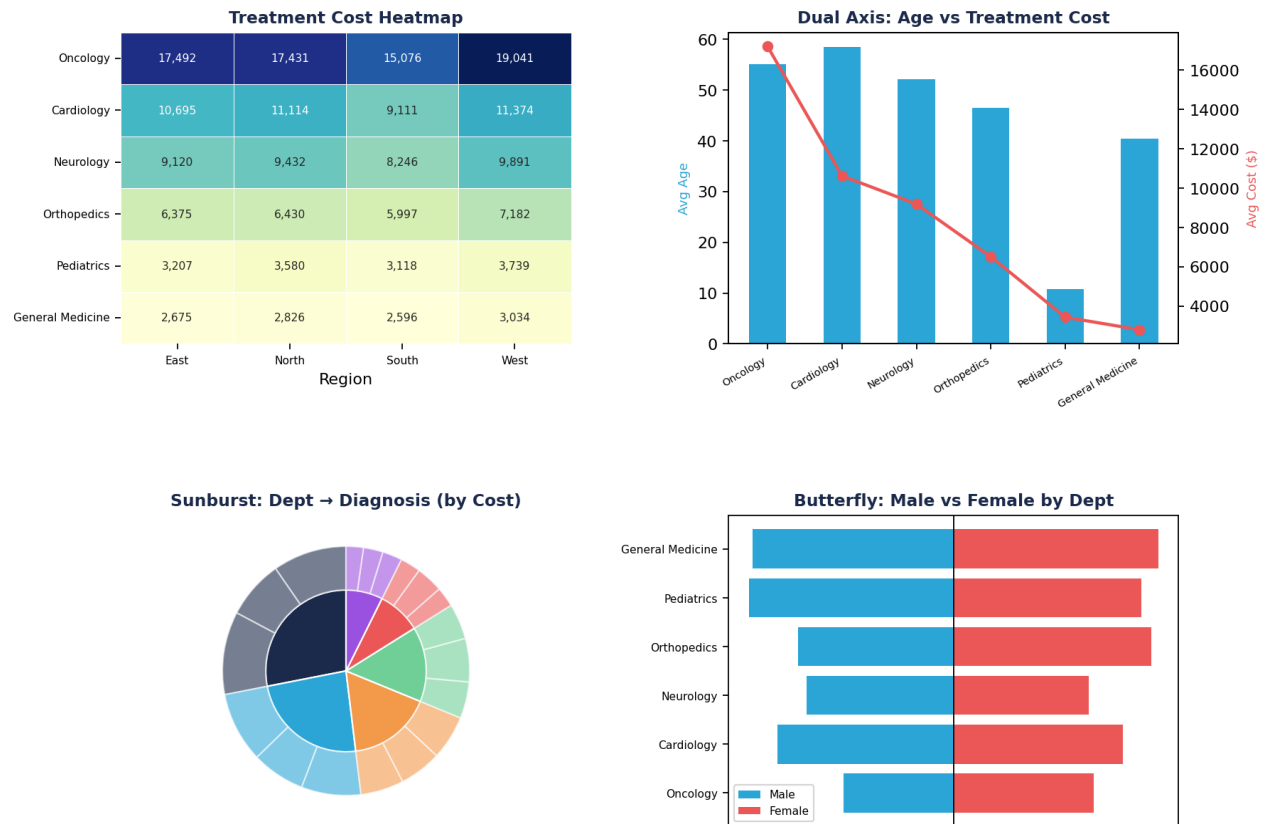


Figure 8: Healthcare Executive Dashboard combining Heatmap, Dual Axis, Sunburst, and Butterfly views

This single view lets hospital administrators see cost distribution by region, the age/cost relationship, the diagnosis-level cost hierarchy, and gender balance by department all at once — and drill into any specific department or region with a single click across every chart.